Vimarsh Dwivedi

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Education

Bachelor of Technology in Computer Science Engineering May 2022 – May 2026
Vellore Institute of Technology, Bhopal, India CGPA: 8.32/10.0

Core Competencies

Programming Languages: JavaScript, Python, HTML5, CSS3, SQL, C++, Java
Frontend Development: React.js, Responsive Design, Modern CSS, DOM Manipulation, ES6+
Backend Development: Node.js, Express.js, RESTful APIs, JWT Authentication, MongoDB, MySQL, PostgreSQL, SQLite
Developer Tools & Technologies: Git, Docker, AWS, GCP, CI/CD, Flask, FastAPI, Postman, npm
Machine Learning: TensorFlow, PyTorch, Scikit-learn, LangChain, LangGraph, RAG, NLP

Professional Experience

Coherent Market Insights, Pune September 2025 – Present
AI Engineer

- Built a single platform that makes AI coding tools **safer and cheaper** to use, combining **secret scanning**, smart routing across **286 skills** and **68 expert agents**, a **live code-intelligence graph**, and an **LLM cost-optimization gateway** — all accessible through **one API key** and working across **5 editors** (Cursor, Claude Code, Windsurf, VS Code, and Zed).
- Created a **pre-push guardrail system** that automatically checks code before it is pushed — blocking **hardcoded secrets, API keys, and risky AI-generated changes** from ever reaching the repository, so unsafe code is caught on the developer's machine instead of in production.
- Replaced the gateway's **in-memory state** (which was wiped on every restart) with **permanent multi-tenant storage** for cache and spend/savings records on both **SQLite** and **PostgreSQL**; tested with **500 simultaneous writes** and confirmed **zero lost updates**.
- Found and fixed a **major billing loophole** where unknown LLM models were charged as **$0**, letting users quietly exceed spending limits; reworked the pricing layer to **block unknown models by default**, with a simple config option to add new ones.
- Fixed a bug in the **caching and model-routing** interaction that overreported cost savings by up to **96%**, and added **automated regression tests** so savings numbers stay accurate going forward.
- Built a **benchmark suite** on a real **214,280-line, 162-file** codebase showing **92.3% fewer tokens** (**13.0× less context**) for code-intelligence queries versus grep-then-read, and **70.6% fewer tokens** across a realistic **16-prompt** coding session (never below **65.5%**, clearing the required **60%** threshold).

CodeFeast, Remote July 2025 – September 2025
Software Engineer

- Built Python-based automation and assertion frameworks across WhatsApp, Gmail, and Google Calendar to validate real-world communication and scheduling flows for LLM agents.
- Developed 15+ tailored evaluation scripts that expanded the internal LLM testing framework, improving scenario coverage across 25+ reasoning benchmarks through dynamic validation pipelines.

Nexyom, Remote March 2025 – July 2025
Data Science Intern

- Designed end-to-end RAG and Graph RAG pipelines that enhanced context recall and semantic matching for large document sets, enabling more accurate information retrieval at scale.
- Implemented a document forgery detection framework using LLMs and metadata analysis, achieving 93.24% accuracy in identifying tampered legal and medical documents across 2,000+ test cases.

Technical Projects

Multi-Modal Real Estate Price Prediction | *Python, TensorFlow, CNN, LSTM, Flask* GitHub

- Developed a hybrid CNN+LSTM neural network processing 53,000+ property records with multimodal data, achieving 92% property valuation accuracy.
- Optimized the feature engineering pipeline with dimensionality reduction on 25+ variables, increasing model generalization by 18% and reducing training time by 35%.
- Engineered a containerized Flask API with RESTful endpoints serving real-time valuation predictions, supporting 100+ concurrent requests with horizontal scaling.

Medical GenAI Assistant | *Python, LangChain, LangGraph, Hugging Face, Qdrant* GitHub

- Architected a multi-tool medical agent using LangGraph and RAG with dynamic query routing across Qdrant vector DB (top-3 retrieval), PubMed API, and a custom web scraper, driving a 38% boost in response accuracy.
- Designed an automated ingestion pipeline chunking medical texts into 8k-token segments with 768-dimensional embeddings, enabling sub-second query responses at a 0.78 cosine similarity threshold.